Warsaw Lean 4 users: intro

Paweł Balawender

Tuesday, 27th of October, 2026

Section 1

Plan

Plan

- Intro + organisation
- What was happening in summer?
- What is happening next month?
- Direction 1: AI4Math
- Direction 2: Software verification

Section 2

Intro + organization

Key info

- Website with announcements: https://warsaw-lean4-users.github.io/
- GitHub org: https://github.com/warsaw-lean4-users
- Meetings: monthly. Probable next (approximate):
  - 24th of November
  - then 15th of December.
- Communication: Lean Zulip: Warsaw location topic.
- We're not a "koło naukowe" yet
- We don't have pizza sponsorship yet

Goal of the meetings

- **Help each other start and finish projects centered around verification, by:**
  - staying up to date with ongoing events, announcements
  - feedback for project ideas
  - creating space for people of different backgrounds to learn about verification
- This talk will mostly be a recap of what was/is happening. Future talks are free to have different format

My calendar project

- I've gathered all the events that were at least remotely interesting
- Some focus on student stuff, including mentorships and volunteering opportunities
- Please check out! Even small contributions are welcome :)
- https://ruplet.com/ai4math-calendar

Section 3

Recap of last months: general events

June, July (selected)

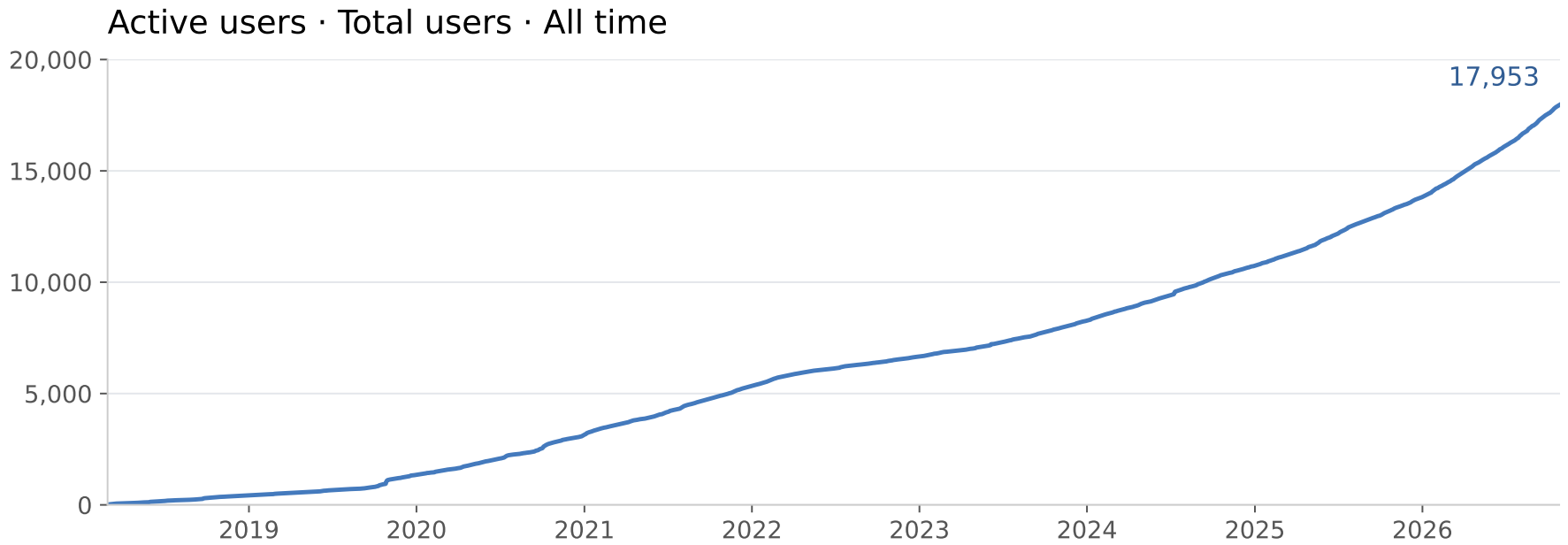
- 29 June–3 July: ECOOP 2026 - Brussels; including UNSOUND
- 11 July: AI4Math workshop at ICML 2026
- 13–29 July: Federated Logic Conference (FLoC 2026), Lisbon, including:
  - Conference on Interactive Theorem Proving (ITP)
  - Conference on Computer Aided Verification (CAV)
  - Conference on Automated Reasoning (IJCAR)
  - Symposium on Logic in Computer Science (LICS)
  - Computer Security Foundations Symposium (CSF)
  - competitions for SAT, SMT and QBF solvers
- FLoC is once every 4 yers! Next one in 2030

August, September (selected)

- 24–29 August ICFP 2026 + affiliated events
- 7–11 September: Proof Society 2026
- 15–18 September: FMCAD 2026
- 21–25 September: CICM 2026
- 4–9 October: SPLASH 2026
- Full calendar: https://ruplet.com/ai4math-calendar

Zulip statistics

17 953 total users. Only 1671 users active in the last 15 days.



Source: https://leanprover.zulipchat.com/stats

What was talked about on Zulip?

- 1. (1718 messahes, 272 reactions) Physlib: Yukawa sector of the Standard Model
- 2. (264 messages, 724 reactions) Counterexample to the Collatz Conjecture: soundness bug.
- 3. (195 messages, 708 reactions) Anthropic's FLT formalization (will discuss later)

Section 4

What is happening this and next month?

October

- 26—30 Oct: Bristol Lean week
- 24—29 Oct: EMNLP 2026: Budapest.
- 31 Oct: ELLIS applications are closing for PhD students
- 31 Oct: Mathlib reviewer bootcamp application is closing

November

- 6 Nov: EACL '27 Student Research Workshop (student research paper / phd thesis feedback)
- 7 Nov: Conference on Software Verification in Lean SVIL '26 Fall - can join online to watch streams
- 11 Nov: POPL '27 student volunteers: registration + meals; travel / lodging excluded. Zulip call.
- 16 Nov: NAACL '27 Student Research Workshop
- 30 Nov: AAAI '27 scholar / volunteer applications opens

Section 5

AI4Math

Discussion of OpenAI: Navier–Stokes + the massive GitHub dump

- 8 September: Navier Stokes annonuncement; paper; Lean dump.
- 6 October: Navier–Stokes lost in translation, arXiv:2610.08144: paper / Lean mismatch; $m + 4$ vs. $m + 5$ derivatives. No claim that the written proof is incorrect.
- Lean Zulip discussion
- 6 October: OpenAI GitHub dump
- 16 October: How to Formalize All of Mathematics; post by Jeremy Avigad, Kevin Buzzard, Johan Commelin, Rémy Degenne, Kim Morrison, Oliver Nash and Adam Topaz

Anthropic + community responses

- 4 September: Anthropic's Fermat's Last Theorem formalization
- 4 September: Kevin Buzzard's comment

Kevin Buzzard's FLT project continues. Their goal is to produce mathlib contributions + a human-readable modern proof

AI4Math funding: Polish project with Renaissaince Philantropy grant

- Renaissance Philantropy is a major grant-given in the field of AI4Math
- This year, a Polish team lead by Bartosz Naskręcki (UAM / PW) received one of the grants
- Congratulations!
- links: Award list, Agentic Approach to Computer Algebra Systems

Section 6

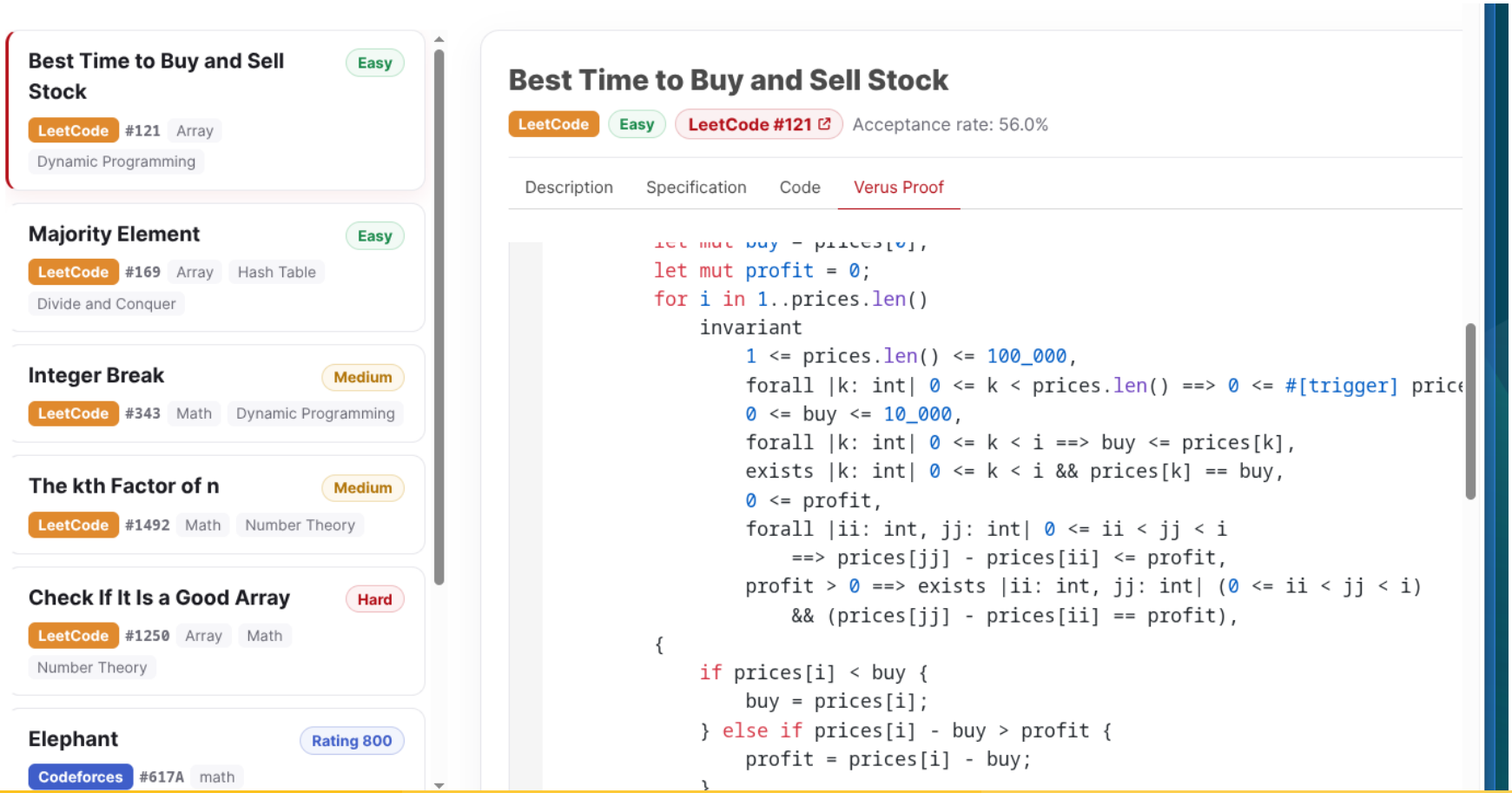
Software verification

Verification of imperative programs

- Until recently, state of the art was using Why3 or Dafny
- Dafny's SMT solvers tend to get stuck; not a production-ready tool
- there was a proof mode proposal for Dafny (link) with Amazon Research funding, but it seems to not be there yet
- important well-established project: Aeneas (verifying Rust using Lean 4); ICFP 2022

Verus

- At NeurIPS 2026, VeriContest will be presented: benchmark for solving LeetCode problems in Rust **with correctness proof in Verus** (Verus is a tool from around 2024)
- Looks good (link)



Verifying Rust standard library

- in May 2026, NASA Formal Methods Symposium (NFM 2026) took place, and the paper "Verifying the Rust Standard Library" was published link
- they accept many tools as verification, including Verus
- FMCAD 2026: a few talks about verifying real Rust; 18 September "Verifying a Memory Allocator for Rust"

Signal Shot

- April 20, 2026 at SVIL workshop in Paris
- verification of the whole Signal communicator app + the protocol
- https://www.beneficialaifoundation.org/signal-shot
- supported by Beneficial AI Foundation - hint towards financing from AI Safety direction
- other AI Safety grant-giver worth mentioning: Coefficient Giving link

Velvet

- CAV, July 2026: “Velvet: A Foundational Multi-Modal Verifier for Imperative Programs in Lean”
- https://velvetprover.dev/

```
    do
      if res[mind]! < res[mind - 1]! then
        let tmp := res[mind]!
        res := res.set! mind res[mind - 1]!
        res := res.set! (mind - 1) tmp
      mind := mind - 1
    n := n + 1
  return res

prove_correct insertionSort by
  velvet_vcgen [insertionSort] with try finish
  case inner_sorted =>
    intro i j hij hj_bound hj
    by_cases hj_m : j = mind
    · subst hj_m
      rcases Nat.lt_or_eq_of_le hij with hi | rfl
      · exact Int.le_trans
```

Cryptography

- CSF 2026 (part of FLoC): Dziembowski, Fabiański, Micciancio, Stefański, Computationally-Sound Symbolic Cryptography in Lean
- and more works by Warsaw crypto group - but IMO it's worth dedicating a separate meeting for their works

Cybersecurity: HORIZON-ECCC call

- there was a call that could fund some formal verification “Security testing methodologies, including formal verification approaches and AI-driven security testing methodologies;”
- call by European Cybersecurity Centre, ECCC
- closed on September 15, but it suggests that cybersecurity-related funding is possible
- link

Cybersecurity: ECCC-CYBERAI call (CPPC)

- there is an open call, but further from formal verification (until 14 January 2027)
- “Cybersecure tools, technologies and services relying on AI”
- https://www.gov.pl/web/cppc/nabory-2026-z-obszaru-cyberbezpieczenstwo

FIRST TEAM FENG

- until 20 November, FIRST TEAM FENG is open (FNP)
- the call looks very broad - perhaps someone knows more about this program?